

**Q 1-Open Command Prompt or Terminal and use the ping command to test connectivity to [www.spotify.com](https://www.spotify.com). Note the number of packets sent, received, and the average response time.**

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
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C:\Users\ratho>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=15ms
Reply from 2a04:4e42:31::810: time=39ms
Reply from 2a04:4e42:31::810: time=22ms
Reply from 2a04:4e42:31::810: time=35ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 39ms, Average = 27ms

C:\Users\ratho>|
```

The connectivity to [www.spotify.com](https://www.spotify.com) was successfully tested using the ping command. 4 packets were sent and 4 packets were received, with 0% packet loss. The average response time was 27 ms.

**Q 2-Use the traceroute (tracert on Windows, traceroute on Linux/Mac) command to trace the path from your computer to [www.flipkart.com](https://www.flipkart.com). List the number of hops and identify the IP address of the first and last hop.**

```
C:\Users\ratho>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [64:ff9b::b856:f8b1]
over a maximum of 30 hops:

  1  <1 ms    <1 ms    <1 ms    2409:40c1:4026:5c60::37
  2  *         *         *         Request timed out.
  3  16 ms    24 ms    28 ms    2405:200:5210:3:3925::1
  4  19 ms    13 ms    9 ms     2405:200:898:3632:62::9
  5  29 ms    22 ms    11 ms    64:ff9b::c0a8:e3c3
  6   6 ms    16 ms    23 ms    64:ff9b::c0a8:f48c
  7  *         *         *         Request timed out.
  8  *         *         *         Request timed out.
  9  *         *         *         Request timed out.
 10  20 ms    25 ms    28 ms    a184-86-248-1.deploy.static.akamaitechnologies.com [64:ff9b::b856:f801]
 11  34 ms    32 ms    17 ms    a184-86-248-177.deploy.static.akamaitechnologies.com [64:ff9b::b856:f8b1]

Trace complete.

C:\Users\ratho>|
```

The traceroute to [www.flipkart.com](https://www.flipkart.com) completed in 11 hops. The first hop was 2409:40c1:4026:5c60::37, and the final hop was 64:ff9b::b856:f8b1. Some intermediate hops showed request timeouts, but the trace successfully reached the destination.

**Q 3-Simulate a connectivity issue by pinging a non-existent website (for example, ping `www.fakezomato12345.com`) and write down the error message you receive.**  
**Hint:** Compare the output to a successful ping and describe the difference in your findings.

```
Command Prompt
C:\Users\ratho>ping www.spotify2y.com
Ping request could not find host www.spotify2y.com. Please check the name and try again.
C:\Users\ratho>
```

The ping was unsuccessful because the specified domain name could not be resolved. Unlike the successful ping to `www.spotify.com`, no reply or packet statistics were received. This indicates a DNS/name resolution problem for the non-existent domain.

**Q 4-Imagine you are unable to access `www.youtube.com` from your system. Use both ping and traceroute commands to diagnose the issue and write a short report describing where the connection fails and what the possible reasons could be.**

```
Command Prompt
Microsoft Windows [Version 10.0.19045.4894]
(c) Microsoft Corporation. All rights reserved.
C:\Users\ratho>ping www.youtube.com

Pinging www.youtube.com [142.250.192.206] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 142.250.192.206:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\ratho>tracert www.youtube.com

Tracing route to youtube-ui.l.google.com [142.250.192.206]
over a maximum of 30 hops:
  0  <1 ms    <1 ms    <1 ms    192.168.1.1
  1  8 ms     7 ms     8 ms     10.10.10.1
  2  14 ms    15 ms    14 ms    100.64.0.1
  3  *        *        *        Request timed out.
  4  *        *        *        Request timed out.
  5  *        *        *        Request timed out.
  6  *        *        *        Request timed out.
  7  *        *        *        Request timed out.
  8  *        *        *        Request timed out.
  9  *        *        *        Request timed out.
 10 *        *        *        Request timed out.

Trace complete.
C:\Users\ratho>
```

Ping to [www.youtube.com](http://www.youtube.com) did not receive replies, and traceroute showed that the connection reached the ISP network before later hops stopped responding. This suggests that the local network is likely functioning, while the failure may be beyond the local router. However, traceroute timeouts alone cannot prove that the connection is actually broken because intermediate routers or YouTube's network may intentionally block traceroute/ICMP traffic.